

ShopCloud - Cloud-Based Secure Web Application

Capstone Project Report (DevSecOps)

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Live URL: <http://15.206.94.233>

GitHub: <https://github.com/aswinkrishnam555/cloud-ecommerce-app>

1. Architecture

- AWS EC2 (Ubuntu 24.04) — Web Server
- Nginx — Reverse Proxy
- Flask (Python) — Backend
- SQLite — Database
- GitHub Actions — CI/CD
- CloudWatch — Monitoring

2. Security

- bcrypt password hashing
- JWT authentication
- UFW Firewall (ports 22, 80, 443)
- Fail2ban brute-force protection
- SSH root login disabled
- Role-based access control

3. Deployment Guide

1. Clone repo from GitHub
2. Install requirements: `pip install -r requirements.txt`
3. Run locally: `python run.py`
4. AWS: EC2 + Nginx + nohup
5. CI/CD: auto-deploy on git push

4. Monitoring

- CloudWatch agent installed
- CPU, Memory, Disk metrics
- App logs: `/ecommerce/app`
- Nginx logs: `/var/log/nginx/`

5. Application Features

- User registration and login
- Product listing with cart
- Admin dashboard
- CRUD operations